Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

AI engineer with **9+ years of production software engineering** and deep hands-on ownership of agentic AI systems, RAG pipelines, and LLM evaluation. At Intuit, builds and operates **production-grade AI agents**: agent reasoning loops, dynamic tool calling, MCP integrations, structured output validation (Pydantic), and guardrails for PII filtering and prompt injection defense in regulated financial environments. Built **Leasel** - an 8-agent LangGraph system covering the full FDE brief: document ingestion, vector retrieval, sub-agent delegation, human-in-the-loop escalation, and an LLM-as-judge evaluation harness (93.6% precision across 1,240 test cases). At Entrata (100K+ MAU), collaborated directly with **business end-users** to observe operational workflows, capture domain logic, and ship iterative solutions from prototype to production. Strong in **Python, REST APIs, Docker, AWS, Anthropic Claude**. Daily Claude Code user. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Agent Reasoning Loops & Dynamic Tool Calling | MCP Server Development & API Connectors | RAG Pipelines (Parsing, Embeddings, Vector Indexing) | Human-in-the-Loop (HITL) Escalation | Pydantic / Structured Output Validation | Guardrails: PII Filtering & Prompt Injection Defense | LLM Evaluation & CI/CD Eval Suites | Python / SQL / REST APIs / gRPC | Anthropic Claude / LangGraph / LangChain | AWS / Docker / Kubernetes / Terraform / GitHub Actions | PHI / PII / HIPAA / Regulated Data Compliance | Direct Business Collaboration & Iterative Delivery

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - AI Agents & Production AI Systems - Virginia, USA

- **Build production-grade AI agents** embedded within Line of Business squads: agent reasoning loops, dynamic tool calling, short-term/long-term memory management via LangGraph state, sub-agent delegation, and structured output validation using Pydantic and JSON Schema to guarantee type-safe outputs from foundation models.
- Build and maintain **MCP server integrations** and REST API connectors linking AI agents to enterprise financial systems: ticketing systems, relational databases, message queues, and internal microservices.
- Develop **RAG components**: document parsing, embedding generation, vector indexing (ChromaDB, Pinecone), metadata filtering, and semantic search retrieval for financial document workflows.
- Integrate **real-time guardrail policies**: prompt injection defense, PII filtering/redaction, and input validation around all LLM calls in regulated financial data environments; implement HITL escalation triggers for ambiguous model outputs.
- Partner with C12-equivalent engineers to build **automated evaluation suites** (LLM-as-judge + golden datasets) integrated into GitHub Actions CI/CD; collocate with business end-users to capture domain logic and iterate rapidly on agent prototypes.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users · Python, Java, AWS, Terraform

- **Collaborated directly with business end-users** (operations teams, onboarding leads) to observe day-to-day workflows, capture nuanced domain logic, and translate requirements into production solutions - same pattern as embedded FDE delivery.
- Built **resilient async processing systems** (RabbitMQ): retry logic with exponential backoff, error handling, and fallback routines across high-volume data pipelines processing 1.6M+ records.
- Owned **PHI/PII regulated data infrastructure**: AWS S3 migration with Libsodium encryption, HIPAA-compliant access controls, audit logging - meeting enterprise compliance requirements in a regulated environment.
- Owned **Terraform IaC and GitHub Actions CI/CD**: Docker microservices (50% better scalability, 30% faster deployments); full AWS stack (EC2, RDS, S3, Lambda, SQS, IAM).
- Reviewed 300+ pull requests, coached junior engineers; managed 20+ concurrent workstreams across 120+ Agile sprints.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - **Pune, India**

- Built production Python/Java/PostgreSQL modules for regulated housing workflows; delivered large-scale data import pipelines (1.6M records) with validation and reconciliation.

PROJECTS

Leasel - 8-Agent AI System (Full FDE Brief)

Python · LangGraph · FastAPI · MCP · RAG · HITL · Pydantic · LLM-as-Judge

End-to-end agentic AI system independently scoped and shipped: **agent reasoning loops, dynamic tool calling, sub-agent delegation** (LangGraph), RAG pipeline (OCR ingestion, vector indexing, semantic retrieval), **Pydantic structured output validation**, guardrails routing eligibility decisions away from the LLM, and **HITL escalation** for low-confidence outputs. LLM-as-judge evaluation harness with golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases integrated into CI/CD. Covers every core technical requirement in the Citi FDE brief.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - Automated LLM Evaluation Suite

Python · FastAPI · Anthropic Claude · 5 LLM Providers · CI/CD · Golden Datasets

15-endpoint FastAPI service integrating **Anthropic Claude** and four other LLM providers. Evaluates factual accuracy, hallucination rate, retrieval quality, and compliance via LLM-as-judge. **Automated regression testbed with golden datasets** wired into CI/CD - the same evaluation infrastructure Citi's C12 engineering pattern calls for.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017